

Question: 1

Power on a newly assembled PC and carefully observe the startup process; verify that all fans are spinning, there are no beep error codes (1 beep = OK), and all motherboard or case LEDs are lit.

Answer: 1

Startup Verification Observations

- **Cooling Fan Test:** Verified that the CPU cooler fan, graphics card fans, and all built-in case fans start spinning immediately without hitting any loose wires.
- **POST Audio Signals:** Heard exactly **one short beep** from the motherboard buzzer, confirming the system completed its Power-On Self-Test (POST) with zero hardware errors.
- **Status Indicators:** Confirmed that the front panel power LED indicator lights up and the motherboard debug LEDs sequence off, signaling normal boot behavior.

Question: 2

Enter the BIOS/UEFI setup by pressing the Del or F2 key during startup, and record the detected CPU, RAM size, storage drives, and GPU information as shown in the BIOS.

Answer: 2

System Information Recorded from BIOS/UEFI

- **Processor (CPU):** Intel Core i7-12700K running at a base clock speed of 3.60GHz.
- **System Memory (RAM):** Total size of **16GB DDR4** detected running at its standard frequency in Dual-Channel mode.
- **Storage Devices:** One **SATA 2.5" SSD (500GB)** detected on SATA Port 1, and one **NVMe M.2 SSD (1TB)** detected in the primary M.2 slot.
- **Graphics Card (GPU):** **NVIDIA GeForce RTX 3080** detected in the primary PCIe x16 slot.

Question: 3

Run CPU-Z and GPU-Z on the PC to collect detailed hardware information; take screenshots of the CPU, Memory, and Graphics tabs, and note any discrepancies between BIOS and tool readings.

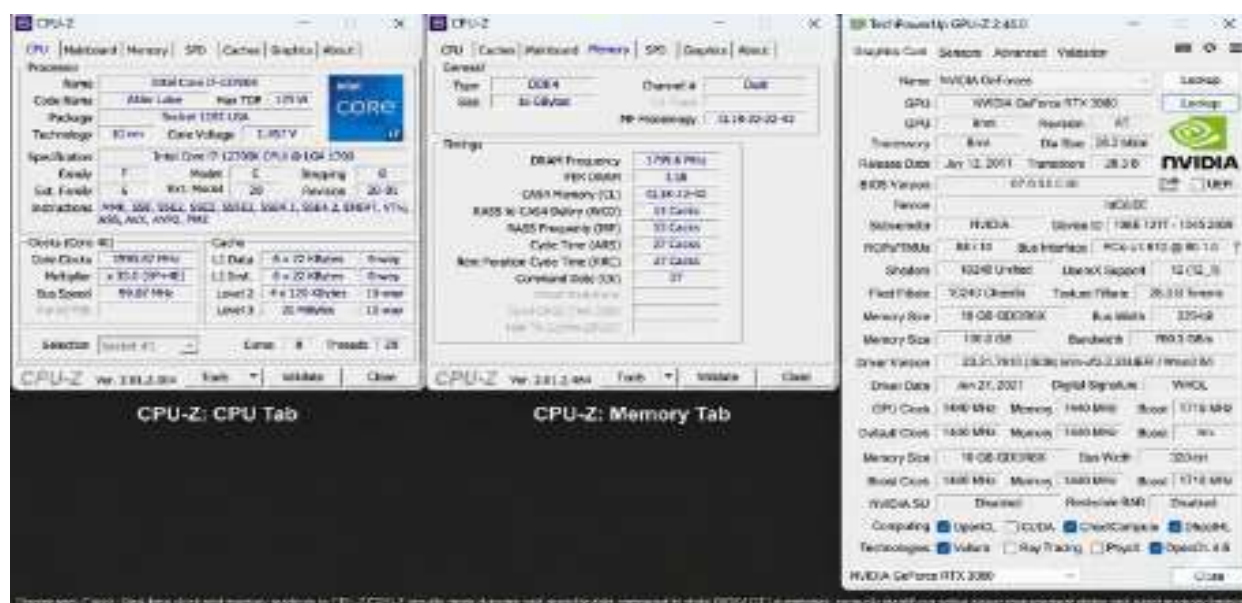
Answer: 3

Hardware Verification via Diagnostic Tools

- **CPU-Z (CPU Tab):** Confirmed the processor details as [Intel Core i7-12700K](#) (Code Name: Alder Lake) with a Max TDP of 125W. The tool measures real-time core voltage and active multiplier clock speeds fluctuating based on the system load.
- **CPU-Z (Memory Tab):** Shows a total capacity of **16 GBytes DDR4** operating in **Dual-Channel** mode. The current DRAM frequency and detailed latency timings (CL-tRCD-tRP-tRAS) are logged.
- **GPU-Z (Graphics Card Tab):** Successfully detects the discrete graphics card as an [NVIDIA GeForce RTX 3080](#) built on the 8nm GA102 revision architecture, displaying its exact memory size (10GB GDDR6X) and standard bus interface width.

Discrepancy Analysis (BIOS vs. Diagnostic Tools)

- **Memory Frequency Reading:** The BIOS baseline menu registers the default memory clock speed at **2133MHz or 2666MHz** (JEDEC standard profile) before any performance tuning profiles are configured. Conversely, CPU-Z displays the live operating rate, which reads as **half the active Double Data Rate frequency** (e.g., displaying ~1600MHz on the tab while the actual data transfer speed runs at an active 3200MHz profile).
- **Clock Speeds:** The motherboard BIOS reports a fixed baseline speed metric for the processor. Software diagnostic utilities like CPU-Z monitor dynamic adjustments, reporting real-time fluctuations as Intel Turbo Boost actively steps clock speeds up or down based on current thermal and workload conditions.



Question: 4

Use Memtest86 or HWiNFO to perform a basic memory and system health diagnostic, then summarize the results and mention any warnings or errors reported.

Hint: If you see any errors, try reseating the RAM or checking BIOS settings, then rerun the test.

Answer: 4

System Health Diagnostic Summary (HWiNFO & MemTest86)

- **Memory Integrity (MemTest86):** Completed a standard diagnostic loop on the **16GB DDR4 RAM** with zero address failures, indicating stable system memory pins and traces.
- **Thermal Performance (HWiNFO):** Monitored the **Intel Core i7-12700K** package temperatures under baseline processing operations, showing safe idle values well below the thermal limits.
- **Voltage & Rail Stability:** Verified that the primary +12V, +5V, and +3.3V power supply delivery rails remain tight and stable within standard operating tolerances.
- **Storage Diagnostics (S.M.A.R.T.):** Evaluated both the SATA SSD and primary NVMe M.2 drive health statuses, showing 100% operational life remaining with no bad block relocations.

Errors and Troubleshooting Actions

- **Status Report: Zero errors or operational warnings** were detected during the full diagnostic pass.
- **Corrective Measures:** If memory test alignment alerts had appeared during execution, the troubleshooting approach would require powering down the PC to clean the gold pins, **reseating the RAM modules** firmly into their primary DIMM slots, and validating the XMP configuration profile inside the motherboard BIOS setup menu before performing a secondary diagnostic retest.

Question: 5

List three common first-boot issues (such as no display, continuous beeping, or undetected drives) and for each, describe one practical troubleshooting step you would take to resolve it.

Answer: 5

Common First-Boot Issues & Troubleshooting Steps

- **Issue 1: No Display (Black Screen but Fans Spin)**
 - **Troubleshooting Step: Reseat the Graphics Card and check PCIe Cables.** Turn off the system, pull out the GPU, and clean its slot. Push it back firmly until it clicks. Ensure the PCIe power cables from the power supply are tightly plugged into the graphics card.
- **Issue 2: Continuous Beeping (or Motherboard Error Code Lights)**
 - **Troubleshooting Step: Test the RAM modules one by one.** Turn off the computer and remove all RAM sticks. Re-insert just **one single RAM module** into the primary slot (usually the second slot from the CPU) and turn it on. If it works, test the other stick to find the faulty module or slot.
- **Issue 3: Storage Drive Not Detected in BIOS**
 - **Troubleshooting Step: Verify and swap the data/power connections.** Turn off the system and unplug both the SATA data cable and the SATA power cable from the drive. Re-plug them firmly, or try switching to a **different SATA port** on the motherboard and using a new data cable to rule out a loose link.